

Novel view synthesis for underwater scenes with Gaussian splat fields and physically-based water modeling

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ABSTRACT

We present a neural rendering framework for few shot underwater scene synthesis that enhances 3D Gaussian Splat Fields with medium aware rendering. Our method represents each scene using anisotropic Gaussian primitives whose shape, appearance, and opacity parameters are predicted by integrating triplane feature embeddings with lightweight neural fields. To model underwater light transport, we design a differentiable volumetric renderer that learns view dependent medium parameters. We further ensure geometric fidelity through a depth consistency loss that reduces artifacts caused by sparse or degraded observations. Experiments on real world underwater benchmarks demonstrate that our approach consistently outperforms underwater view synthesis methods in quantitative accuracy and perceptual quality, especially under sparse input conditions.

1. Introduction

Synthesizing novel photorealistic views of a 3D scene from sparse input images remains a core challenge in computer vision and graphics. Although neural rendering methods [1,2], have achieved impressive performance in free space environments, their effectiveness drops dramatically in underwater scenes. Underwater imaging suffers from complex light transport phenomena, such as wavelength dependent attenuation, backscattering, and refraction, which violate standard rendering assumptions. Furthermore, the availability of multiview data in underwater environments is often limited due to hardware constraints and environmental occlusions, making few shot synthesis particularly challenging.

Neural Radiance Fields (NeRF) [1] and its extensions [3,4] generate highly realistic renderings by representing scenes as continuous volumetric neural fields. However, these methods rely on dense multiview data, require heavy computation, and struggle with long range volumetric sampling. Underwater turbidity, scattering, and reduced visibility further amplify these problems by degrading the quality and density of the data. In contrast, 3D Gaussian Splatting [2] offers a more efficient solution by representing scenes as collections of learnable anisotropic Gaussian primitives, rendered through a scalable rasterization pipeline. However, the original 3DGS framework was designed for clear air scenes and fails to model underwater specific optical effects such as depth-dependent attenuation, backscattering, and refractive distortion.

Recent efforts have attempted to extend neural rendering into underwater domains with physically grounded models. For example, SeaThruNeRF [5] incorporates depth aware attenuation into the NeRF formulation, improving underwater realism. WaterSplat [6] enhances 3DGS with scattering aware rendering, and SeaSplat [7] introduces layered volumetric splatting to simulate wave induced distortions and visibility loss. Despite these advances, existing approaches still face key limitations: they require dense multiview capture for stable training, rely on static or manually defined underwater parameters instead of adaptive learned ones, and lack explicit directional regularization. As a result, these methods often overfit and amplify noise, especially in low texture or highly scattered regions.

We address these challenges with a new framework for few shot underwater scene synthesis that unifies medium aware rendering and regularized 3D Gaussian Splatting. Our method jointly learns scene geometry and underwater appearance by predicting adaptive anisotropic Gaussian primitives conditioned on sparse views, while explicitly modeling the physical properties of the medium. A triplane feature generator encodes limited input images into rich directional embeddings that guide both splat deformation and neural field predictions of geometry and appearance. In addition, a neural medium module learns view dependent attenuation and scattering coefficients, enabling medium adaptive water property modeling. To maintain geometric consistency under sparse supervision, we introduce a depth consistency loss based on

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expected rendered depths. By integrating physically accurate rendering with learned scene representation, our approach significantly improves few shot neural rendering in realistic underwater conditions. Our main contributions are as follows:

- We introduce a novel framework for few shot underwater scene synthesis that extends 3D Gaussian Splatting with physically grounded, medium aware rendering.
- We employ a triplane based feature generator and neural fields to predict adaptive Gaussian primitives from sparse inputs, incorporating a differentiable underwater image formation model that accounts for attenuation, scattering, and backscattering.
- We propose a depth consistency regularization term that strengthens geometric fidelity under sparse view settings.
- Extensive experiments show that our method substantially outperforms baselines on challenging underwater datasets.

2. Related work

2.1. Novel view synthesis

Recent surveys [8,9] provide comprehensive overviews of progress in neural rendering and novel view synthesis. Among these, Neural Radiance Fields (NeRF) [1] represent a major milestone by achieving photorealistic rendering through implicit modeling of volumetric density and view dependent radiance with multilayer perceptrons (MLPs). However, NeRF's reliance on dense ray sampling and repeated MLP evaluations makes both training and inference computationally expensive. To overcome these limitations, several alternative scene representations have emerged. Discrete volumetric methods such as Plenoxels [10] use sparse voxel grids with spherical harmonics to support efficient gradient based optimization. Hybrid representations like Instant-NGP [11] accelerate rendering through hash-encoded positional grids coupled with compact MLPs. Extensions including Nerfies [12] model nonrigid motion via learned deformation fields, while mip-NeRF [13] and mip-NeRF 360 [14] address aliasing through scale-aware positional encoding.

In contrast to implicit volumetric fields, 3D Gaussian Splatting (3DGS) [2] represents a scene using anisotropic Gaussian primitives with learnable positions, covariances, and spherical harmonic coefficients, enabling highly efficient rasterization based rendering. Mip-Splatting [15] improves anti-aliasing via frequency aware integration, VAST-Gaussian [16] scales the representation to large outdoor environments, and SG-SLAM [17] demonstrates real time SLAM with Gaussian primitives. Except for these general algorithms, several works adapt Gaussian Splatting to domain specific or constrained view settings. MonoSplat [18] reconstructs 3D scenes from monocular video using geometric pseudo depth regularization. Recent works introduce photometric regularization for 3D Gaussian Splatting to improve multi-view consistency and geometry stability [19]. FlexGS [20] introduces flexible Gaussian deformation to handle dynamic and restricted viewpoints, and EndoGaussian [21] customizes 3DGS for medical and endoscopic imaging by modeling nonrigid motion, complex illumination, and tubular scene geometry. These domain focused methods improve generalization under strong priors but typically rely on well controlled dynamic setups, ignoring complex light medium interactions. Unlike these efforts, our work targets underwater scene synthesis from sparse static multiview images, where visibility is severely degraded by wavelength-dependent attenuation, backscattering, and refraction. As noted in [8], existing NeRF-based methods struggle to jointly estimate geometry and optical medium properties. Similarly, current Gaussian-based renderers [2] assume clear-air conditions and cannot model volumetric medium effects. To bridge this gap, we introduce a physically grounded extension of 3DGS that integrates learnable medium-aware volumetric rendering, enabling disentangled estimation of underwater photometrically realistic view synthesis.

2.2. Few shot novel view synthesis

Synthesizing novel views from a limited number of input images is a critical yet challenging problem, especially in contexts where dense multiview capture is impractical. Traditional methods such as Multiplane Images (MPIs) [22] model a scene using fronto parallel planes and warp them to novel views via homographies. Recent methods, inspired by NeRF and 3DGS, address the few shot setting with multiple main strategies. The first improves the consistency of the cross view to mitigate the ambiguities arising from the sparse input. DietNeRF [23] regularizes latent features to promote semantic consistency across views, while RegNeRF [24] introduces losses in photometric and geometric consistency to improve stability during training.

The other strategy incorporates depth priors to guide scene reconstruction. DSNeRF [25] utilizes sparse depth maps from SfM pipelines like COLMAP [26] as geometric supervision, and SparseNeRF [27] leverages monocular depth estimates refined via local ranking and smoothness constraints. Recent Gaussian-based methods also extend few shot capabilities: FSGS [28] enables real time view synthesis with Gaussian splatting with dense prior initialization and proximity guided Gaussian unpooling; Chan et al. [29] increase reconstruction quality by densifying Gaussian point clouds from sparse inputs; and He et al. [30] enhance sparse view 3DGS with local depth and semantic regularization. While these approaches improve renderings under sparse view conditions, they assume clear and calibrated environments. In contrast, we focus on the additional challenges of underwater imaging, where scattering, attenuation, and refractive distortion degrade visibility. Our method addresses these complexities by integrating few shot scene reconstruction with medium aware neural rendering, enabling photometrically realistic and geometrically faithful underwater view synthesis.

2.3. Underwater scene restoration

Underwater vision has attracted growing attention in recent years, driven by the increasing demand for reliable perception and reconstruction in marine exploration, robotics, and long term environmental monitoring [31–33]. Traditional underwater image restoration methods largely adapt dehazing techniques from terrestrial vision. Early approaches [34–36] decompose an image into scene radiance, ambient illumination, and transmission, typically assuming wavelength independent attenuation modeled as exponential decay with depth. These assumptions, however, fail to capture the complex spectral properties in underwater environments.

Recent underwater photometric stereo based learning approaches demonstrate that surface normals can be recovered under challenging underwater lighting and scattering [37]. For example, SeaThruNeRF [38, 39] introduces a revised image formation model that accounts for wavelength dependent attenuation and separates direct transmission from backscatter. Although physically accurate, these methods usually depend on external calibration tools such as color charts or depth sensors, which restrict their applicability in real world underwater scenarios. Learning based approaches aim to overcome these limitations by training neural networks to restore degraded underwater images without explicit model inversion. Methods such as WaterGAN [40] and its extensions [41–43] employ supervised or adversarial training [44] to enhance perceptual quality. However, they often fail to disentangle medium effects from scene geometry, limiting generalization across viewpoints and environments. The underwater photometric stereo transformer [45] improves geometric reasoning by reconstructing surface normals under varying illumination, but it still relies on dense image observations.

Recent progress in physically grounded neural rendering offers a more unified solution by jointly modeling scene structure and medium properties. SeaThru-NeRF [5] extends mip-NeRF 360 [14] with wavelength-dependent attenuation and view-aware volumetric modeling for geometry-consistent color correction. WaterSplat [6] enhances 3D Gaussian Splatting (3DGS) with scattering aware rendering, and SeaSplat [7] introduces layered volumetric splatting to simulate wave

induced distortion and depth dependent degradation. Despite these advances, existing approaches still rely on dense multiview inputs for stable optimization and lack explicit geometry regularization, making them prone to noise amplification and overfitting in highly scattering or low texture regions. In contrast, our method targets few shot underwater novel view synthesis by integrating medium aware 3D Gaussian Splatting with learnable transmittance modeling. This formulation explicitly disentangles geometry from medium effects, enabling robust and photometrically consistent rendering even under sparse view conditions.

3. Method

Given a set of calibrated underwater images, our work produces novel views of the scene under arbitrary camera poses. To achieve this, our pipeline integrates four key components: triplane based feature generation, which encodes sparse observations into structured directional embeddings; neural field based splat parameter prediction, which estimates the orientation, scale, color, and opacity of adaptive 3D Gaussian primitives; medium aware underwater rendering, which incorporates attenuation, scattering, and refraction effects along viewing rays; and depth consistency regularization, which enforces geometric coherence across novel viewpoints. An overview of the framework is illustrated in Fig. 1.

3.1. Preliminaries

Following prior work [2], we represent a 3D scene as a set of anisotropic Gaussian primitives $\{G_k\}_{k=1}^K$. Each primitive G_k is defined by a center position $\mathbf{p}_k \in \mathbb{R}^3$, a scaling matrix $\mathbf{S}_k \in \mathbb{R}^{3 \times 3}$, and a rotation matrix $\mathbf{R}_k \in \mathbb{R}^{3 \times 3}$, typically parameterized using a quaternion to ensure numerical stability. The density function of each Gaussian is given by:

$$G_k(\mathbf{x}) = \exp\left(-\frac{1}{2}(\mathbf{x} - \mathbf{p}_k)^\top \boldsymbol{\Sigma}_k^{-1}(\mathbf{x} - \mathbf{p}_k)\right), \quad (1)$$

where the covariance matrix $\boldsymbol{\Sigma}_k \in \mathbb{R}^{3 \times 3}$ is constructed as:

$$\boldsymbol{\Sigma}_k = \mathbf{R}_k \mathbf{S}_k \mathbf{S}_k^\top \mathbf{R}_k^\top. \quad (2)$$

This formulation allows each Gaussian to model an arbitrarily oriented ellipsoid in a 3D space, enabling a compact and expressive representation of complex scene geometry.

3.2. Tri-plane feature generation

Following prior scene representation methods [46], we initialize the scene representation from low dimensional Gaussian noise $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. More specifically, previous work [46,47] takes as input low dimensional Gaussian noise $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and gradually upsamples it to the target image resolution and optimizes the network weights to fit the observed noisy image. Inspired by this concept, we adapt this prior formulation for feature generation in our 3D Gaussian splat representation.

As 3D CNNs are computationally expensive, we employ a lightweight 2D convolutional generator g_θ that produces three axis aligned feature planes (tri-planes) [46]. Each plane encodes features along one coordinate axis, allowing efficient 3D reasoning without volumetric convolution. This stage can be viewed as a splat based variation of the volumetric feature generation used in [1]. Specifically, given a randomly initialized noise vector ϵ , the convolutional network g_θ predicts three orthogonal feature maps $\mathbf{F}$ with resolution $H \times W$:

$$\mathbf{F} = g_\theta(\epsilon) \in \mathbb{R}^{3 \times H \times W \times l}, \quad (3)$$

where l denotes the feature dimension and θ represents the learnable weights of the network. The CNN progressively upsamples the latent noise into structured features, providing a strong implicit regularization prior that captures geometric coherence across views.

For each initial 3D point $\mathbf{p}_k$ (typically obtained from sparse SfM reconstruction [48]), we project its coordinates onto each of the three feature planes and bilinearly interpolate the corresponding feature values. The interpolated features are concatenated into a unified embedding $\mathbf{f}_k \in \mathbb{R}^{3l}$, which encodes both geometric and appearance priors.

The combined feature $\mathbf{f}_k$ and the corresponding point $\mathbf{p}_k$ are then passed through a deformation network f_Θ , which refines the point location:

$$\hat{\mathbf{p}}_k = f_\Theta(\mathbf{p}_k, \mathbf{f}_k), \quad (4)$$

This allows the system to adjust the location of the splat in a data driven manner, enabling better alignment with the geometry inferred from the sparse underwater views. The refined point and its features are further processed by a set of compact neural fields $\{f_{scale}, f_{rot}, f_{opacity}, f_{color}\}$ to predict the parameters of anisotropic Gaussian splats, including scale, rotation, opacity, and view dependent color [46]. Instead of using spherical harmonics for color modeling, we directly regress view dependent color components, improving efficiency and stability.

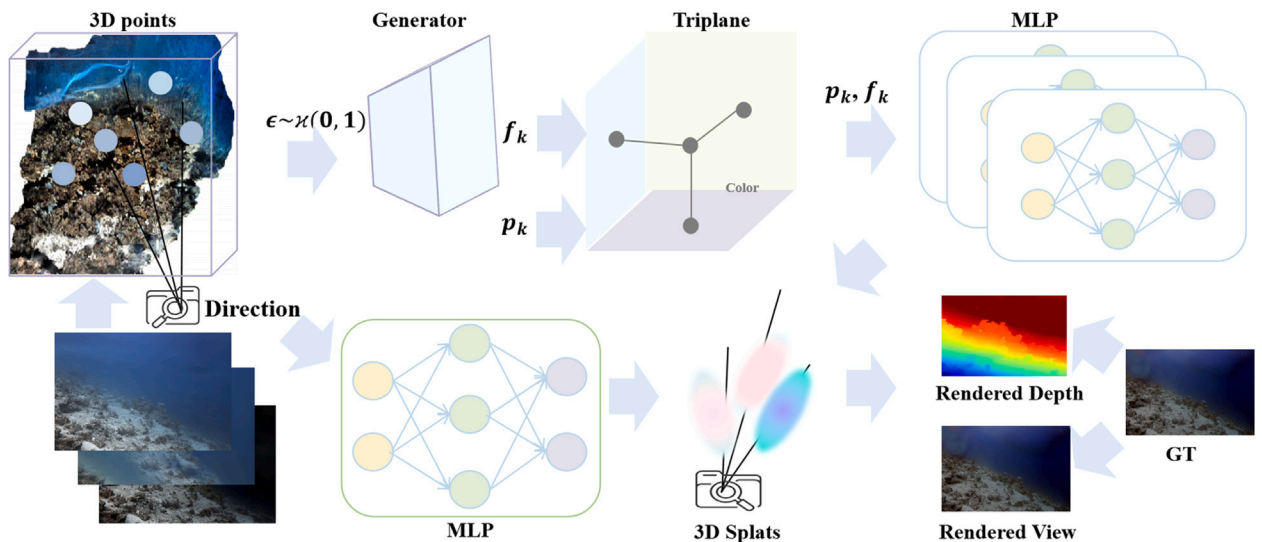


Fig. 1. Overview of our proposed framework. Our neural rendering pipeline for few shot underwater scene synthesis integrates triplane based feature generation, underwater rendering and depth consistency regularization to effectively handle challenging underwater scenarios.

3.3. Neural field based splat prediction

Given the refined position $\hat{\mathbf{p}}_k$ and the characteristic vector $\mathbf{f}_k$, we predict the parameters of each 3D Gaussian splat using lightweight neural fields. We use compact MLPs to regress the splat parameters from $(\hat{\mathbf{p}}_k, \mathbf{f}_k)$ [46], including the orientation, scale, opacity, and color:

$$\begin{aligned}\Sigma_k &= f_{\text{scale}}(\hat{\mathbf{p}}_k, \mathbf{f}_k), & \mathbf{q}_k &= f_{\text{rot}}(\hat{\mathbf{p}}_k, \mathbf{f}_k), \\ \alpha_k &= f_{\text{opacity}}(\hat{\mathbf{p}}_k, \mathbf{f}_k), & \mathbf{c}_k &= f_{\text{color}}(\hat{\mathbf{p}}_k, \mathbf{f}_k, \mathbf{v}),\end{aligned}\quad (5)$$

where Σ_k denotes the anisotropic covariance, $\mathbf{q}_k$ the unit quaternion for rotation, α_k the opacity and $\mathbf{c}_k$ the view dependent color under viewpoint $\mathbf{v}$. Unlike [2], our work does not use spherical harmonics for color modeling but instead regresses the appearance directly using viewpoint conditioned features, which is beneficial in underwater domains where light directionality and color shift are significant.

3.4. Medium aware rendering

In underwater environments, image formation is governed by complex light transport phenomena, including direction dependent attenuation and back scattering. The observed pixel intensity I can be modeled as a combination of a direct transmission term and a back scattered light term [5]:

$$I = O \cdot e^{-\beta^D(\mathbf{v}_D) \cdot z} + B^\infty \cdot \left(1 - e^{-\beta^B(\mathbf{v}_B) \cdot z}\right), \quad (6)$$

where O denotes the radiance of the scene in the absence of the medium, z is the depth along the viewing ray, and $\beta^D(\mathbf{v}_D)$, $\beta^B(\mathbf{v}_B)$ are direction-dependent attenuation coefficients for direct and backscattered light, respectively. B^∞ represents the asymptotic color of the backscatter at infinite depth.

Standard 3D Gaussian Splatting (3DGS) [2] assumes free space radiative transfer and thus does not account for medium-induced effects such as wavelength dependent attenuation, scattering, and refraction. To address this limitation, we incorporate a physically grounded underwater image formation model, inspired by revised formulations in [5,38], into the 3DGS framework. Specifically, We parameterize the camera ray as $\mathbf{r}(s) = \mathbf{o} + s\mathbf{d}$, where $\mathbf{o} \in \mathbb{R}^3$ denotes the ray origin, $\mathbf{d} \in \mathbb{R}^3$ is the unit viewing direction for the pixel, and $s \geq 0$ is the distance along the ray. We model the expected color $C(\mathbf{r})$ along a camera ray $\mathbf{r}(s)$ by integrating radiance from the camera to infinity:

$$C(\mathbf{r}) = \int_0^\infty T(s) \left[\sigma_{\text{obj}}(s) c_{\text{obj}}(s) + \sigma_{\text{med}}(s) c_{\text{med}}(s) \right] ds, \quad (7)$$

where $s \geq 0$ is the continuous ray parameter $\sigma_{\text{obj}}(s)$ and $\sigma_{\text{med}}(s)$ denote the (effective) densities of the scene Gaussians and the medium, and $c_{\text{obj}}(s)$, $c_{\text{med}}(s)$ are their corresponding radiance (RGB) values. The transmittance $T(s)$ models the cumulative extinction up to depth s :

$$T(s) = \exp\left(-\int_0^s [\sigma_{\text{obj}}(u) + \sigma_{\text{med}}(u)] du\right). \quad (8)$$

To make the rendering tractable, we discretize the integral along the ray using the ordered set of Gaussians intersected by it. Let $\{s_i\}_{i=1}^N$ be the sorted depths of these Gaussians (i.e., discretized sample locations of the continuous variable s), and define the discrete step length $\Delta s_i \triangleq s_i - s_{i-1}$. With this discretization, the integral measure ds and the discrete step Δs_i are linked via $\int_{s_{i-1}}^{s_i} f(s) ds \approx f(s_i) \Delta s_i$. For the i -th Gaussian centered at depth s_i , the direct object contribution is

$$C_i^{\text{obj}} = T_i \cdot (1 - \exp(-\alpha_i)) \cdot c_i, \quad (9)$$

where T_i is the accumulated transmittance up to s_i , α_i is the opacity-related parameter of the i -th Gaussian, and c_i is its emitted color.

In our discrete form, $T_i = \prod_{j=1}^{i-1} \exp(-\alpha_j - \sigma_{\text{attn}} \Delta s_j)$ with σ_{attn} the attenuation coefficient. The backscatter contribution between adjacent Gaussians at depths s_{i-1} and s_i is

$$C_i^{\text{med}} = T_{i-1} \cdot c_{\text{med}} \cdot [\exp(-\sigma_{\text{bs}} s_{i-1}) - \exp(-\sigma_{\text{bs}} s_i)], \quad (10)$$

and a terminal backscatter term captures contributions beyond the last Gaussian:

$$C_\infty^{\text{med}} = T_N \cdot c_{\text{med}} \cdot \exp(-\sigma_{\text{bs}} s_N). \quad (11)$$

where N denotes the number of Gaussians contributing to the current pixel ray after rasterization, and $\{s_i\}_{i=1}^N$ are their depths sorted in ascending order.

The final rendered color $C(\mathbf{r})$ is the sum of object and medium contributions along a viewing ray $\mathbf{r}$, and it is decomposed into several additive components that account for interactions with both objects and the surrounding medium:

$$C(\mathbf{r}) = \sum_{i=1}^N C_i^{\text{obj}} + \sum_{i=1}^N C_i^{\text{med}} + C_\infty^{\text{med}}. \quad (12)$$

To capture directionally varying water properties, we introduce a learnable neural medium model f_ϕ , which predicts the object attenuation σ_{attn} , the medium backscatter σ_{bs} , and the medium color $\mathbf{c}_{\text{med}}$ from the ray direction $\mathbf{d}$, encoded using sinusoidal positional encoding [1]:

$$(\sigma_{\text{attn}}, \sigma_{\text{bs}}, \mathbf{c}_{\text{med}}) = f_\phi(\mathbf{d}). \quad (13)$$

This formulation enables directionally varying water properties while maintaining differentiability. By integrating explicit 3D Gaussian geometry with a differentiable, medium aware rendering process, our framework faithfully models wavelength dependent attenuation, haze, and backscattering, achieving photometrically realistic novel view synthesis.

3.5. Joint optimization

In this work, all modules, including generator, deformation MLP, and neural fields, are jointly optimized. This end to end training ensures that the triplane features remain geometrically consistent, view-coherent, and physically grounded even under sparse-view supervision. Throughout the training process, we optimize our framework end to end using gradient based backpropagation. The total loss function $\mathcal{L}$ consists of a reconstruction term and a depth regularization term:

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \lambda_d \mathcal{L}_{\text{depth}}, \quad (14)$$

where $\mathcal{L}_{\text{rec}}$ measures the photometric error between rendered and ground truth RGB images, and $\mathcal{L}_{\text{depth}}$ enforces depth consistency to guide accurate geometry reconstruction. The weighting parameter λ_d is set to 1.0×10^{-2} in our experiments.

We adopt $\mathcal{L}_{\text{rec}}$ from the original 3D Gaussian Splatting (3DGS) framework [2], which compares rendered and ground truth images using a loss ℓ_1 . Although this loss encourages photometric consistency, it is insufficient to guarantee accurate geometry in sparse view settings. To address this, we introduce a depth regularization term defined as:

$$\mathcal{L}_{\text{depth}} = \|W \odot (D_r - D_t)\|_1, \quad (15)$$

where D_r and D_t denote the rendered and proxy depth maps (from COLMAP when available), $W \in [0, 1]^{H \times W}$ is a per-pixel confidence map that indicates whether the proxy depth D_t is available at pixel p . We set $W(p)$ to 1 if the depth exists, otherwise set it to 0. The rendered depth D_r is computed as a weighted expectation along each camera ray using:

$$D_r = \frac{1}{\sum_{i=1}^N T(p_i) \alpha(p_i)} \sum_{i=1}^N T(p_i) \alpha(p_i) d_i, \quad (16)$$

where d_i is the center depth of the i -th Gaussian along the ray, $\alpha(p_i)$ is its opacity, and $T(p_i)$ is the accumulated transmittance. This formulation emphasizes contributions from high opacity, front facing Gaussians

and attenuates distant or transparent ones, helping enforce structural accuracy in the reconstruction.

4. Experiments

We evaluated our method on the task of novel view synthesis from sparse multi-view underwater images. Extensive experiments were conducted on two real world benchmarks: the SeaThru-NeRF dataset [5] and the Saltpond dataset [7]. These data sets represent diverse underwater conditions and imaging configurations, allowing a comprehensive assessment of the generalization and robustness of our model.

4.1. Implementation details

We implemented our framework in PyTorch with CUDA acceleration, and all experiments were conducted on a single NVIDIA A100 GPU. The feature extraction architectures follow [46], including the CNN generator g_θ , which produces three axis-aligned feature planes $\mathbf{F} \in \mathbb{R}^{3 \times 160 \times 160 \times 16}$, and multiple MLP modules, which keep same layers and size (5-6 layers, 64–128 neurons) as [46]. We encode the medium using a spherical harmonic (SH) encoding [6,50] and a three layer MLP with 128 hidden neurons and activations. We apply a Sigmoid activation to produce c_{med} , and Softplus activations for σ_{attn} and σ_{bs} . All components, including the triplane feature generator and neural field heads are trained jointly in an end to end manner. We use the Adam optimizer with weight decay of 5×10^{-4} and a cosine learning rate schedule. Each model is trained for around 30k iterations. The initial Gaussian point cloud is seeded from the sparse SfM reconstruction (COLMAP). During training, triplane features are updated continuously with Gaussian parameters and medium coefficients enabling adaptive refinement.

We evaluated all methods using three standard metrics: Peak Signal to Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM) [51], and Learned Perceptual Image Patch Similarity (LPIPS) [52]. Higher PSNR and SSIM values and lower LPIPS values indicate better synthesis quality.

4.2. Dataset

4.2.1. SeaThru-NeRF

The SeaThru-NeRF dataset [5] contains real world underwater scenes captured across four locations: IUI3 Red Sea, Curaçao, Japanese Gardens (Red Sea), and Panama. The scenes consist of 29, 20, 20 and 18 images, respectively, and camera poses are estimated using COLMAP [26]. In the full view experiments, we follow the evaluation protocol of [5,6] to split the training and testing set, among which 25, 17, 17 and 15 images are used for training and the rest 4, 3, 3 and 3 are used for validation. To evaluate performance under sparse view settings, we train all methods with 3 and 6 uniformly sampled views per scene, using the rest for testing.

4.2.2. Saltpond

The Saltpond dataset [7] consists of over 260 downward facing seafloor images captured by a remotely operated underwater vehicle at varying altitudes. In contrast to SeaThru-NeRF's forward-looking configuration, Saltpond emphasizes bottom-facing views with variable scene to camera distances. Camera poses are obtained using COLMAP [48]. We adopt the official train and test splits for full view settings [7]. To evaluate robustness under sparse views, we train all methods using only 3 or 6 uniformly sampled input views per scene, holding out the remaining images for testing.

4.3. SeaThruNeRF dataset

To comprehensively assess the performance of our method under varying levels of supervision, we conduct experiments using 3, 6, and full input views. To ensure reproducibility and fair evaluation, all COLMAP reconstructions and intrinsic parameters were identical to those used in prior works such as SeaSplat [7]. Our baseline includes NeRF-based methods [39] and 3DGS based method [2,7,49,53]. As reported in Table 1, our method consistently achieves the highest accuracy across all evaluation metrics in the full view setting when sufficient data is available. More importantly, as shown in Tables 2 and 3, our approach

Table 1

Quantitative evaluation on the SeaThru-NeRF dataset on full training views. The darker the better (PSNR↑/SSIM↑/LPIPS↓).

Dataset/Metric	Curasao			IUI3 Red Sea			J.G. Red Sea			Panama			Average		
Method	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS
SeaThru-NeRF [39]	25.91	0.79	0.30	30.19	0.87	0.21	21.84	0.77	0.25	27.85	0.83	0.22	26.45	0.82	0.24
3DGS [2]	28.01	0.88	0.21	21.11	0.81	0.29	21.49	0.85	0.22	29.64	0.90	0.17	25.50	0.87	0.21
DepthRGS[49]	28.73	0.89	0.20	23.08	0.86	0.22	21.65	0.83	0.27	30.01	0.91	0.16	25.87	0.87	0.21
SeaSplat [7]	30.30	0.90	0.19	26.67	0.87	0.21	22.70	0.87	0.18	28.76	0.90	0.15	27.11	0.89	0.18
Ours	31.45	0.92	0.19	30.58	0.90	0.19	25.04	0.89	0.17	31.09	0.92	0.13	29.54	0.91	0.17

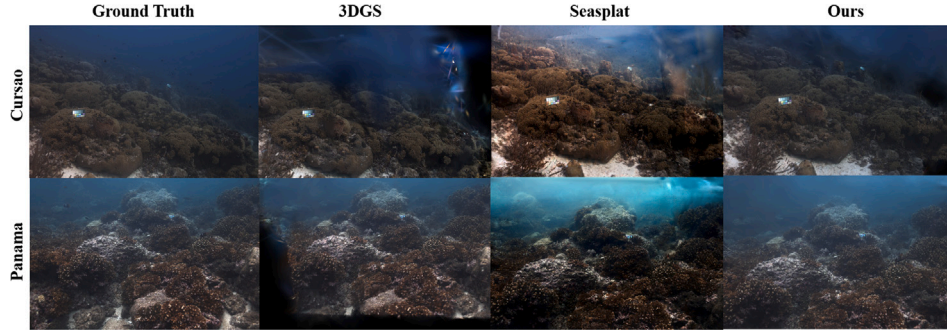
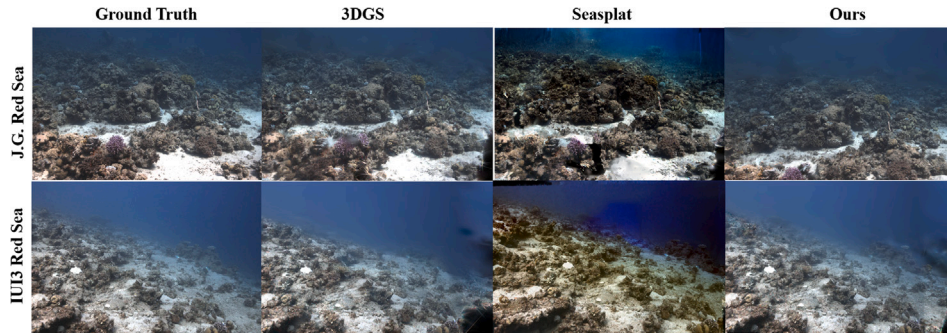
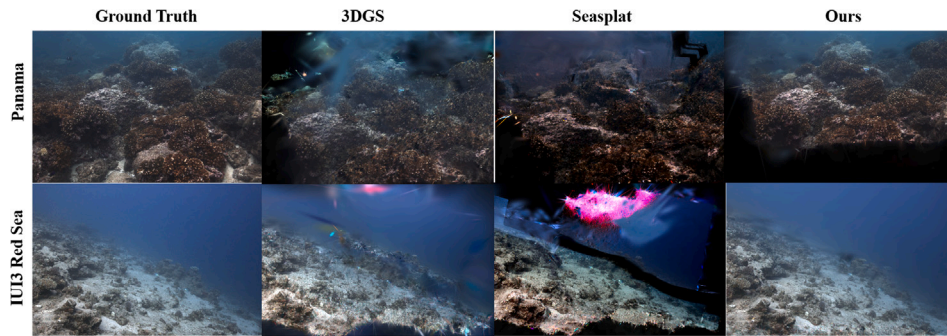
Table 2

Quantitative evaluation using the SeaThru-NeRF dataset on 6 training views. The darker the better (PSNR↑/SSIM↑/LPIPS↓).

Dataset/Metric	Curasao			IUI3 Red Sea			J.G. Red Sea			Panama			Average		
Method	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS
SeaThru-NeRF [39]	18.43	0.56	0.57	17.25	0.58	0.50	14.32	0.52	0.59	18.90	0.67	0.53	17.22	0.58	0.54
3DGS [2]	22.97	0.73	0.37	21.84	0.69	0.34	20.36	0.72	0.32	23.61	0.77	0.38	22.19	0.72	0.35
DepthRGS [49]	23.01	0.75	0.27	22.32	0.69	0.32	20.61	0.70	0.29	23.89	0.71	0.34	22.45	0.71	0.31
SeaSplat [7]	20.53	0.63	0.33	16.81	0.59	0.36	17.52	0.64	0.31	19.62	0.79	0.34	21.29	0.71	0.31
Ours	25.41	0.79	0.24	23.62	0.72	0.31	22.86	0.79	0.25	24.66	0.80	0.29	24.14	0.78	0.27

Table 3Quantitative evaluation on the SeaThru-NeRF dataset with 3 training views. The darker the better (PSNR $\uparrow$ /SSIM $\uparrow$ /LPIPS $\downarrow$).

Dataset/Metric	Curasao			IUI3 Red Sea			J.G. Red Sea			Panama			Average		
Method	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS
SeaThru-NeRF [39]	16.00	0.54	0.62	13.90	0.55	0.57	12.67	0.49	0.61	18.52	0.65	0.59	15.49	0.55	0.59
3DGS [2]	21.36	0.67	0.38	18.59	0.61	0.39	18.23	0.64	0.35	19.71	0.62	0.39	19.48	0.64	0.37
DepthRGS [49]	21.55	0.69	0.27	20.72	0.63	0.37	18.59	0.65	0.31	20.78	0.67	0.35	20.41	0.66	0.33
SeaSplat [7]	19.02	0.53	0.40	14.20	0.52	0.49	16.70	0.56	0.43	18.89	0.67	0.39	17.20	0.57	0.43
Ours	25.12	0.71	0.26	23.16	0.70	0.31	19.16	0.67	0.29	21.93	0.70	0.32	22.34	0.69	0.29

**Fig. 2.** Qualitative comparison of the rendered novel views on the Seathru-NeRF dataset [5] using full training views.**Fig. 3.** Qualitative comparison of the rendered novel views on the Seathru-NeRF dataset [5] using 6 training views.**Fig. 4.** Qualitative comparison of the rendered novel views on the Seathru-NeRF dataset [5] using 3 training views.

maintains a clear performance advantage over existing baselines in challenging sparse view scenarios.

In addition to quantitative performance, Figs. 2–4 visualize side by side comparisons of the rendered results and ground truth. While baselines such as 3DGS and SeaSplat exhibit structural distortions and haze, our visual improvements directly validate the impact of the proposed

depth consistency regularization and medium-aware rendering. The synthesized novel views produced by our model exhibit high visual realism, with well preserved structural details and minimal artifacts. These results validate the practical applicability of our framework in real world underwater conditions where dense view coverage is often infeasible.

4.4. Saltpond dataset

To further demonstrate the generalization capability of our approach, we conduct experiments on the Saltpond dataset. Table 4 reports

Table 4

Quantitative evaluation on the Saltpond dataset on full training views. The darker the better.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
3DGS [2]	27.10	0.75	0.29
MipSplat[53]	27.32	0.76	0.26
Seasplat [7]	27.47	0.75	0.25
Ours	27.72	0.76	0.24

Table 5

Quantitative evaluation with the Saltpond dataset on 6 training views. The darker the better.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
3DGS [2]	22.03	0.54	0.32
MipSplat [53]	22.10	0.55	0.31
Seasplat [7]	15.68	0.48	0.31
Ours	22.79	0.56	0.30

Table 6

Quantitative evaluation on the Saltpond dataset with 3 training views. The darker the better.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
3DGS [2]	19.10	0.76	0.35
MipSplat[53]	19.20	0.77	0.34
Seasplat [7]	14.63	0.41	0.37
Ours	19.49	0.78	0.33

the results under the full view training setting, including 3DGS [2], MipSplat [53], Seasplat [7] and ours. Our method achieves the highest PSNR and SSIM scores among compared approaches, confirming its robustness under dense supervision. We also observed that the Saltpond dataset contains clearer water and lower turbidity than SeaThru-NeRF, leading to smaller relative gains in the full view case. Nevertheless, our model consistently achieves superior performance under sparse view settings (6 or 3 input images), as summarized in Tables 5 and 6.

Qualitative comparisons under the 6 input view setting are provided in Fig. 5. Compared with 3DGS and SeaSplat, our reconstructions exhibit sharper edges and fewer color shifts, indicating effective compensation for wavelength-dependent attenuation. Despite the severe limitation in input views, the proposed method maintains high fidelity in both structure and appearance, demonstrating strong adaptability to diverse underwater scenes.

4.5. Ablation study

To evaluate the effectiveness of each proposed component, we perform a series of ablation experiments on the SeaThru-NeRF validation set. Table 7 reports quantitative results obtained by progressively enabling three key modules: (1) the triplane feature representation, which enhances medium expressiveness under limited views; (2) the medium aware rendering module, which models direction-dependent light attenuation and backscattering; and (3) the depth regularization loss ($\mathcal{L}_d$), which enforces geometric consistency across views. Each variant is evaluated using PSNR, SSIM, and LPIPS metrics to measure reconstruction fidelity and perceptual quality. As shown in Fig. 6, removing any of these modules leads to visible artifacts or degraded sharpness. When all

Table 7

Ablation Study on SeaThru-NeRF.

Tri-plane Feature	Medium Rendering	$\mathcal{L}_d$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
			22.19	0.72	0.35
✓			22.91	0.72	0.29
	✓		23.18	0.75	0.28
		✓	23.67	0.77	0.28
✓	✓	✓	24.14	0.78	0.27

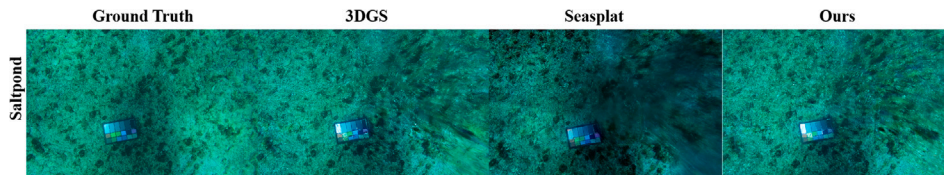


Fig. 5. Qualitative comparison of the rendered novel views on the Saltpond dataset [6] using 6 training views.

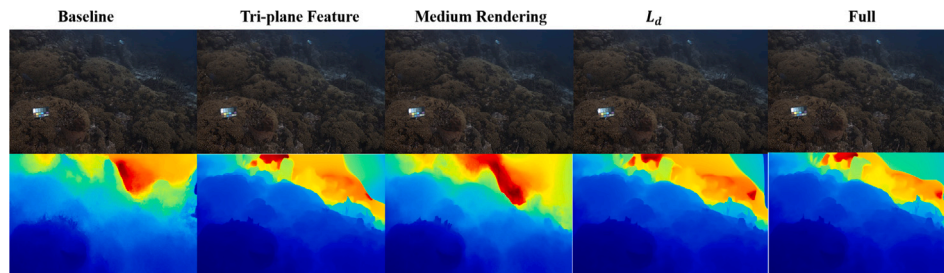


Fig. 6. Ablation study on the SeathruNeRF dataset [6] using 6 training views.

three components are activated, the complete model achieves the highest scores across all metrics, clearly demonstrating their complementary benefits. This verifies that jointly modeling underwater light physics, and geometric constraints is essential for robust and photorealistic few shot underwater novel view synthesis.

5. Conclusion

We presented a novel framework for few shot underwater novel view synthesis that enhances 3D Gaussian Splatting with physically grounded volumetric rendering. By combining directionally adaptive Gaussian primitives with learnable medium aware rendering, our method produces photorealistic results even under sparse view supervision. Experiments on real world underwater datasets show that our approach consistently surpasses underwater view synthesis baselines, establishing a new state of the art in underwater rendering. In future work, we could explore extending this framework to dynamic underwater scenes with temporal coherence, adapting it to diverse aquatic conditions, and integrating it with real time perception systems for marine robotics and underwater environmental monitoring.

CRedit authorship contribution statement

Qian Li: Writing – review & editing, Writing – original draft, Visualization, Methodology. **Rao Fu:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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